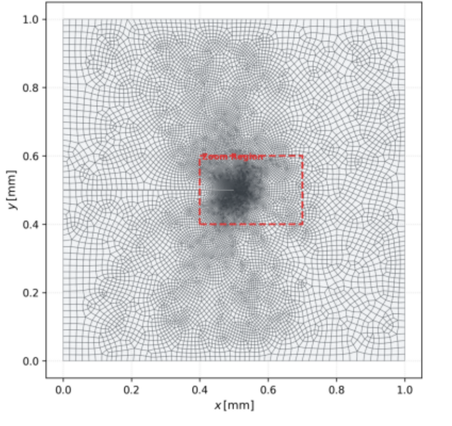
Method A Adaptive Remeshing Framework — Five-Geometry Benchmark Verification
Unified Native Engine Evaluation | Status: FIVE_CASE_METHOD_A_FRAMEWORK_PARTIALLY_VALIDATED

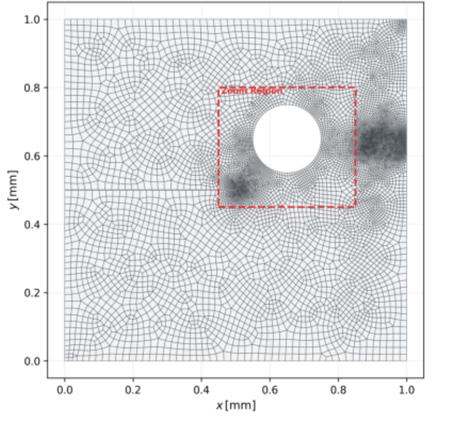
Case 1: Mode-I Edge-Cracked Plate (Sec. 4.1)
Adapted: 14,147 el (Target: 13,941 el) | PASS

Final Adapted Physical Mesh: Model B
Physical Elements: 14933 (Quads: 14493, Tris: 440, Nodes: 14895)



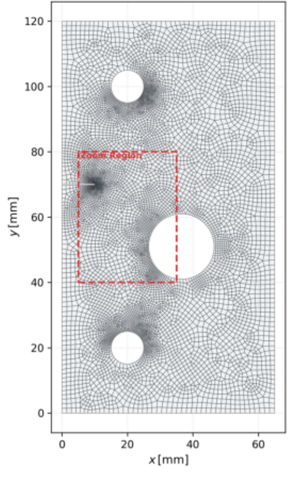
Case 2: Off-Axis Circular Hole Plate (Thesis)
Adapted: 23,965 el (Distal Resolved) | PASS

Final Adapted Physical Mesh: Model C
Physical Elements: 11302 (Quads: 10973, Tris: 329, Nodes: 11375)

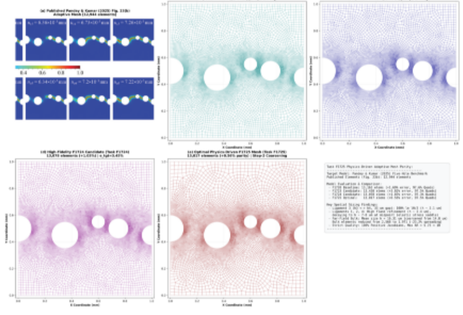


Case 3: Notched Plate with Hole & Fins (Sec. 4.3)
Adapted: 12,284 el | PARTIALLY_VALIDATED (Single Corridor)

Final Adapted Physical Mesh: Model A
Physical Elements: 9432 (Quads: 9132, Tris: 300, Nodes: 9567)



Case 4: Five-Hole Plate (Sec. 4.5)
Adapted: 13,017 el (Target: 12,944 el) | PASS



Case 5: Seven-Hole Plate (Sec. 4.6)
Adapted: 15,708 el (Target: 15,714 el) | PASS

